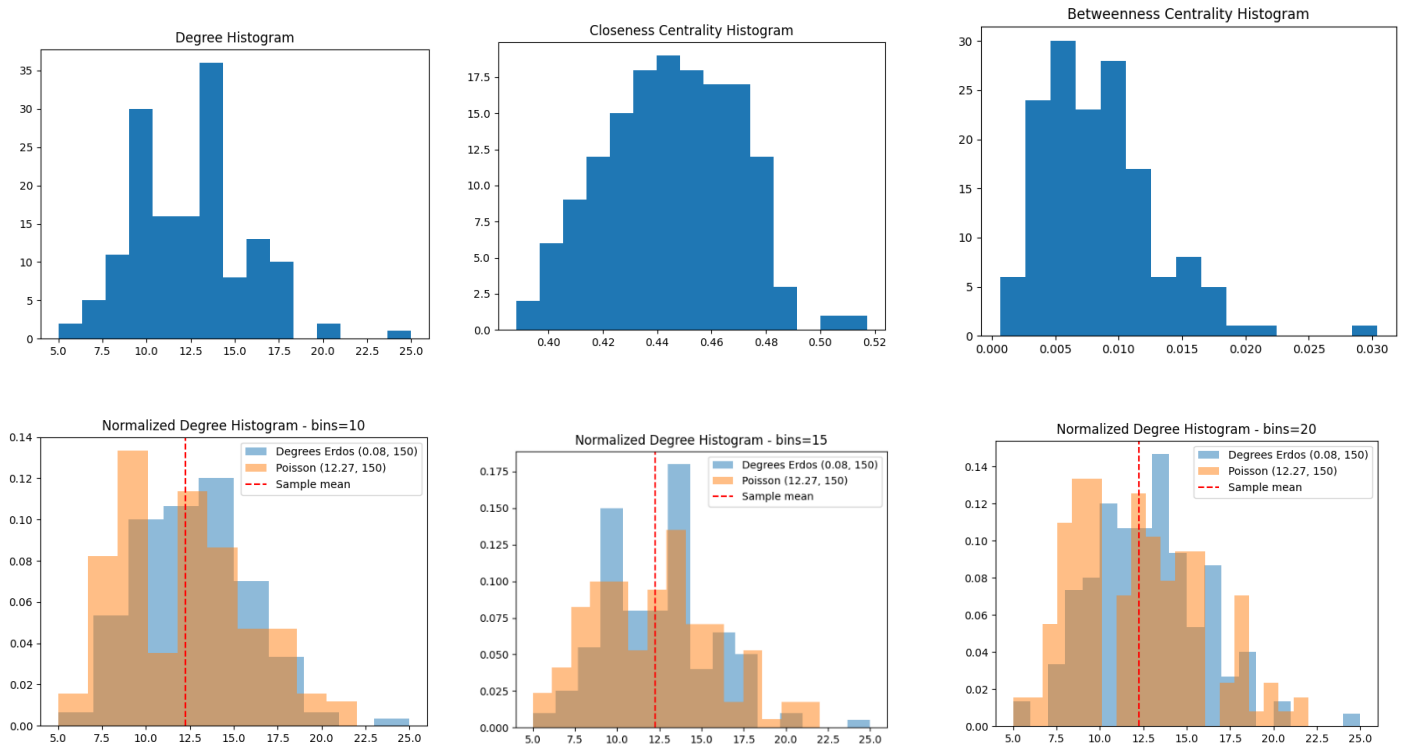
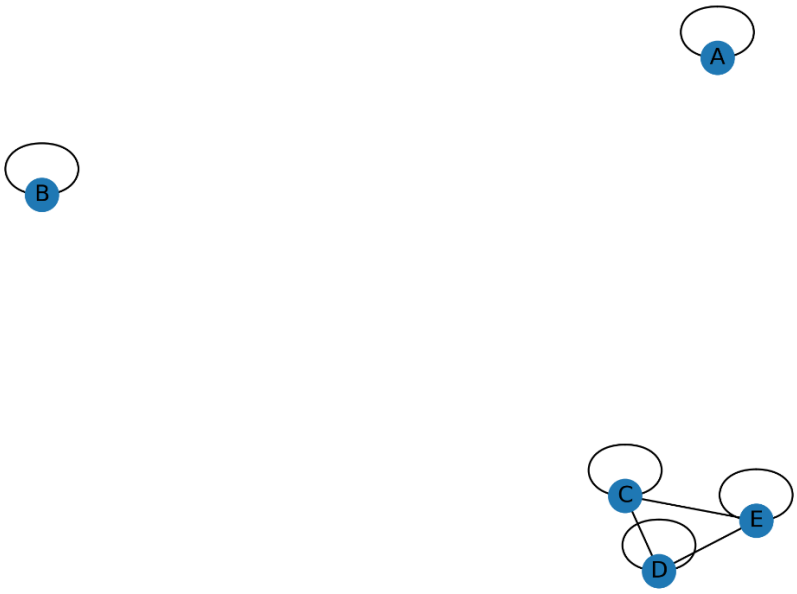
Task 3

1. Generate Erdos-Renyi G(n=150,p=0.08).
2. Analyze:
   - number and size of connected components
   - histograms for: degree value, closeness, and betweenness centralities
   - normalize degree histogram and plot Poisson PMS over it (scipy.stats.poisson)
   - compare for different bin numbers. Q Is the bin size important?
3. Export the network as GraphML file and attributes (centralities, degree) as csv file
4. Check what happens when different 'p' (no need to hand in)



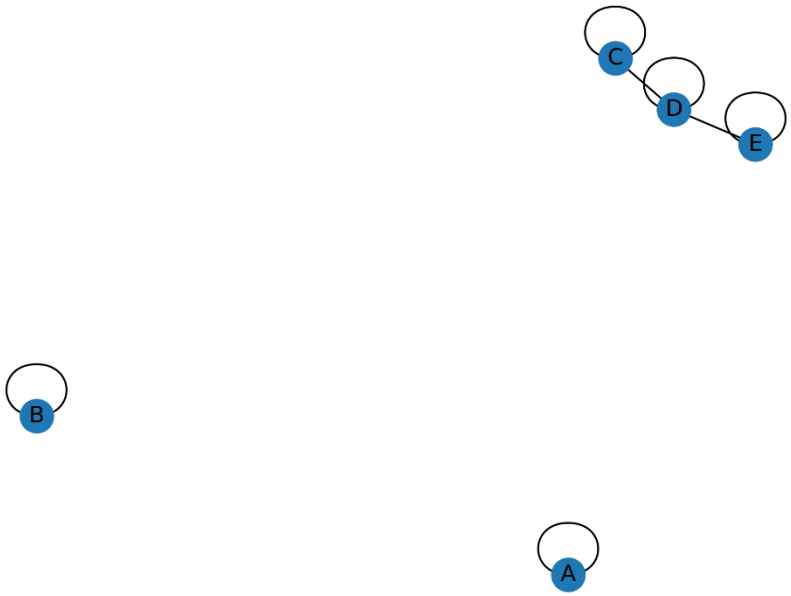
Task 1

Relation if euclidian distance is greater than 11

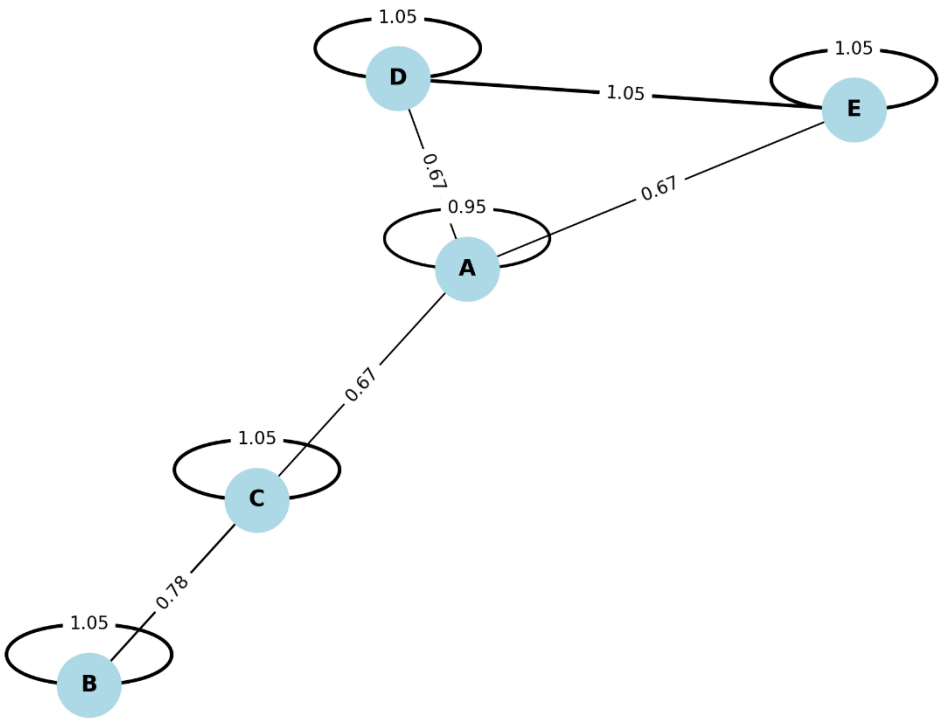


Relation if correlation coefficient is greater than 0.75 (in absolute value). Colored red if positive correlation

Relation if L1 form is smaller than 20



Relation if Mutual Information is greater than 0.65



Data	width	0,2375			width	0,215			
	x1	0,03	0,14	0,18	y1	0,09	0,15		
	x2	0,3	0,39	0,49	y2	0,48			
	x3	0,54	0,65		y3	0,53	0,65	0,71	
	x4	0,91	0,98		y4	0,74	0,8	0,88	0,95

Probability

p(x,y)	x1	x2	x3	x4	marginal p(y)
y1	0	0,1	0	0,1	0,2
y2	0	0	0,1	0	0,1
y3	0,1	0,1	0,1	0	0,3
y4	0,2	0,1	0	0,1	0,4
marginal p(x)	0,3	0,3	0,2	0,2	

p(x)p(y)	x1	x2	x3	x4	marginal p(y)
y1	0,06	0,06	0,04	0,04	0,2
y2	0,03	0,03	0,02	0,02	0,1
y3	0,09	0,09	0,06	0,06	0,3
y4	0,12	0,12	0,08	0,08	0,4
marginal p(x)	0,3	0,3	0,2	0,2	

Independence	NO, because p(x,y) != p(x)p(y)			
Conditional distribution x|y=3	0,33	0,33	0,33	0,00
Conditional distribution y|x=4	0,5	0	0	0,5

Entropy	**2,164**	x1	x2	x3	x4	marginal y	
-1*p(x)*LN(p(x))	y1	0,000	0,230	0,000	0,230	0,322	
	y2	0,000	0,000	0,230	0,000	0,230	
	y3	0,230	0,230	0,230	0,000	0,361	
	y4	0,322	0,230	0,000	0,230	0,367	
	marginal x	0,361	0,361	0,322	0,322		**1,366**
						1,280	

Conditional Entropies

H(X|Y) = X(X,Y)-X(Y)	**0,884**
H(Y|X) = X(X,Y)-X(X)	**0,798**

Mutual information

H(X)+H(Y)-H(X,Y)	**0,4821**		0,4821	x1	x2	x3	x4
Definition	**0,4821**		y1	0,0000	0,0511	0,0000	0,0916
			y2	0,0000	0,0000	0,1609	0,0000
			y3	0,0105	0,0105	0,0511	0,0000
			y4	0,1022	-0,0182	0,0000	0,0223